

Theoretical Quantum Optics: Problem Sheet 5

(for 02.06.2025)

13. Derivation of the master equation (2P)

After the Markov approximation, the general form of the master equation for the reduced system density operator is

$$\dot{\rho}_s(t) = -\frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \text{Tr}_B \{ [\tilde{H}_{\text{int}}(t), [\tilde{H}_{\text{int}}(t_1), \tilde{\rho}_s(t) \otimes \rho_B(0)]] \}, \quad (1)$$

where

$$\tilde{H}_{\text{int}}(t) = \hbar\sqrt{\gamma} [c(t)B^\dagger(t) + c^\dagger(t)B(t)]. \quad (2)$$

The double-commutator gives rise to four different contributions and in the lecture we outlined the evaluation of one of them. Repeat the same calculation for the following two contributions

$$\dot{\rho}_s(t) = \frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \left[\text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) \} + \text{Tr}_B \{ \tilde{H}_{\text{int}}(t_1) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t) \} \right] \quad (3)$$

and show that this will result in the two ‘recycling terms’,

$$\dot{\rho}_s = A_- c \rho_s c^\dagger + A_+ c^\dagger \rho_s c, \quad (4)$$

in the final master equation.

14. Stroboscopic master equation (2P)

Consider a cavity QED system as discussed in the lecture in Sec. 3.3, where Rydberg atoms with states $|g\rangle$ and $|e\rangle$ are interacting resonantly with a single microwave mode. During the time of interaction the Hamiltonian is

$$H_{\text{int}} = \frac{\hbar g}{2} (a\sigma_+ + a^\dagger\sigma_-). \quad (5)$$

Assume that at a time t the cavity is an arbitrary pure state $|\psi_c(t)\rangle$ and a Rydberg atom in state $|g\rangle$ is sent toward the cavity. Therefore, right before the interaction the total state of the system is $|\Psi(t)\rangle = |\psi_c(t)\rangle \otimes |g\rangle$.

- (a) Assume that the Rydberg atom is very fast such that the parameter $\varepsilon = \tau g/2 \ll 1$ is small, but finite. Evaluate the state after the interaction,

$$|\Psi(t + \tau)\rangle = e^{-iH_{\text{int}}\tau/\hbar} |\Psi(t)\rangle, \quad (6)$$

up to order ε^2 .

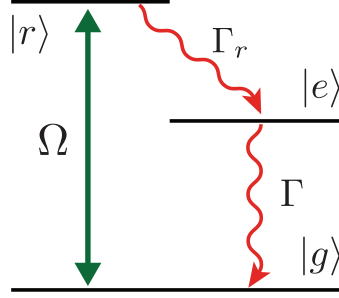


Abbildung 1: Three level atom.

- (b) After the interaction the atom hits a wall and it is lost. Take the trace over the atomic degrees of freedom and calculate the reduced cavity state $\rho_c(t + \tau) = \text{Tr}_{\text{atom}}\{|\Psi(t + \tau)\rangle\langle\Psi(t + \tau)|\}$, again up to order ε^2 .
- (c) This process is repeated and every time interval τ a new atom initialized in state $|g\rangle$ is sent through the cavity. Use the discrete time derivative

$$\dot{\rho}_c(t) = \frac{\rho_c(t + \tau) - \rho_c(t)}{\tau} = \dots \quad (7)$$

to derive an effective master equation for $\rho_c(t)$. Is this master equation of Lindblad form? What is the decay rate?

- (d) How would the final master equation change, if the atoms are initialized in state $|e\rangle$ before passing through the cavity?

Remark: In this example, the environment is a single two-level system with a single frequency. Still the environment is ‘large’, because the atom is renewed all the time and it is Markovian, because after a short interaction time τ , two successive atoms are uncorrelated.

15. Population inversion in a three-level atom (2P)

Consider a three-level atom with an additional state $|r\rangle$ and a level scheme as illustrated in Fig. 1. The master equation for this system is

$$\dot{\rho} = -i[H, \rho] + \frac{\Gamma}{2}\mathcal{D}[\sigma_{ge}]\rho + \frac{\Gamma_r}{2}\mathcal{D}[\sigma_{er}]\rho, \quad (8)$$

where $\sigma_{ij} := |i\rangle\langle j|$ and

$$H = \frac{\Omega}{2}(\sigma_{gr} + \sigma_{rg}). \quad (9)$$

We have also introduced the short notation

$$\mathcal{D}[c]\rho := 2c\rho c^\dagger - c^\dagger c\rho - \rho c^\dagger c. \quad (10)$$

- (a) Write a Python program to simulate this master equation for a three level system and simulate the dynamics of the populations $p_{\eta=g,e,r}(t) = \langle\sigma_{\eta\eta}\rangle(t)$, starting from the ground state, $\rho(0) = |g\rangle\langle g|$. Plot the results for the specific parameters $\Omega = 2$, $\Gamma = 1$ and $\Gamma_r = 0.3$.

- (b) Evaluate the steady state ρ_{ss} as a function of Ω/Γ and for a ratio between the decay rates of $\Gamma_r/\Gamma = 5$. Identify conditions under which a population inversion $p_e(t \rightarrow \infty) > p_g(t \rightarrow \infty)$ can be achieved in the steady state. How can you change the parameters in order to reach an almost ideal inversion, $p_e \rightarrow 1$?

Hint: To get started with this exercise, use the ‘ME_Tutorial.ipynb’ as a guideline.

13. Derivation of the master equation (2P)

After the Markov approximation, the general form of the master equation for the reduced system density operator is

$$\dot{\rho}_s(t) = -\frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \text{Tr}_B \{ [\tilde{H}_{\text{int}}(t), [\tilde{H}_{\text{int}}(t_1), \tilde{\rho}_s(t) \otimes \rho_B(0)]] \}, \quad (1)$$

where

$$\tilde{H}_{\text{int}}(t) = \hbar \sqrt{\gamma} [c(t) B^\dagger(t) + c^\dagger(t) B(t)] \quad (2)$$

The double-commutator gives rise to four different contributions and in the lecture we outlined the evaluation of one of them. Repeat the same calculation for the following two contributions

$$\dot{\rho}_s(t) = \frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \left[\text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) \} + \text{Tr}_B \{ \tilde{H}_{\text{int}}(t_1) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t) \} \right] \quad (3)$$

and show that this will result in the two 'recycling terms',

$$\dot{\rho}_s = A_- \rho_s c^\dagger + A_+ c^\dagger \rho_s c, \quad (4)$$

in the final master equation.

$$\int_{-\infty}^t dt_1 \text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) \}$$

$$\tilde{H}_{\text{int}}(t) = \hbar \sqrt{\gamma} [c(t) B^\dagger(t) + c^\dagger(t) B(t)]$$

$$\begin{aligned} \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) &= \hbar^2 \gamma (c(t) B^\dagger(t) + c^\dagger(t) B(t)) \tilde{\rho}_s(t) \otimes \rho_B(0) (c(t_1) B^\dagger(t_1) + c^\dagger(t_1) B(t_1)) \\ &= \hbar^2 \gamma \left(c(t) B^\dagger(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c(t_1) B^\dagger(t_1) + c(t) B^\dagger(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c^\dagger(t_1) B(t_1) \right. \\ &\quad \left. + c^\dagger(t) B(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c(t_1) B^\dagger(t_1) + c^\dagger(t) B(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c^\dagger(t_1) B(t_1) \right) \end{aligned}$$

$$\text{Tr}_B (c(t) B^\dagger(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c(t_1) B^\dagger(t_1)) = \text{Tr}_B c(t) (B^\dagger(t_1) B^\dagger(t) \tilde{\rho}_s(t) \otimes \rho_B(0)) c(t_1)$$

↑
trace cyclic
no

→ 0 because $B^{\dagger 2}$ -terms

same argument here → 0

$$\text{Tr}_B (c(t) B^\dagger(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c^\dagger(t_1) B(t_1)) = c(t) \text{Tr}_B (B(t_1) B^\dagger(t) \rho_B(0)) \tilde{\rho}_s(t) c^\dagger(t_1)$$

$$\text{Tr}_B (c^\dagger(t) B(t) \tilde{\rho}_s(t) \otimes \rho_B(0) c(t_1) B^\dagger(t_1)) = c^\dagger(t) \text{Tr}_B (B^\dagger(t_1) B(t) \rho_B(0)) \tilde{\rho}_s(t) c(t_1)$$

$$\frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \left[\text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) \} + \text{Tr}_B \{ \tilde{H}_{\text{int}}(t_1) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t) \} \right] = \gamma \int_{-\infty}^t dt_1 c(t) \text{Tr}_B (B(t_1) B^\dagger(t) \rho_B(0)) \tilde{\rho}_s(t) c^\dagger(t_1)$$

$$+ \gamma \int_{-\infty}^t dt_1 c^\dagger(t) \text{Tr}_B (B^\dagger(t_1) B(t) \rho_B(0)) \tilde{\rho}_s(t) c(t_1)$$

substitute: $t_1 = t - \tau$ ($\tau = t - t_1$)

$$\begin{aligned} &\Rightarrow \gamma \int_0^\infty d\tau \text{Tr}_B (B(t-\tau) B^\dagger(t) \rho_B(0)) c(t) \tilde{\rho}_s(t) c^\dagger(t-\tau) \\ &= \gamma \int_0^\infty d\tau \text{Tr}_B (B(t-\tau) B^\dagger(t) \rho_B(0)) e^{-i\omega_s \tau} c(t) \tilde{\rho}_s(t) c^\dagger(t-\tau) \end{aligned}$$

$c(t) = e^{i\omega_s t / \hbar}$
 analogue for $c^\dagger$
 $c^\dagger = e^{-i\omega_s t / \hbar} = e^{-i\omega_s \tau} c^\dagger(t-\tau)$

$$\begin{aligned} \tilde{\rho}_s(t) &= \tau \int_0^\infty d\tau \operatorname{Tr}_B (B^\dagger(t-\tau) B(t) \rho_B(0)) c^\dagger(t) \tilde{\rho}_s(t) c(t-\tau) \\ &= \gamma \left(\int_0^\infty d\tau \operatorname{Tr}_B (B^\dagger(t-\tau) B(t) \rho_B(0) e^{i\omega_s \tau}) \right) c^\dagger(t) \tilde{\rho}_s(t) c(t) \end{aligned}$$

now

$$\begin{aligned} &\operatorname{Tr}_B \{ \tilde{H}_{\text{int}}(t_1) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t) \} \quad (\\ &= \gamma \underbrace{\int_{-\infty}^t dt_1 c(t_1) \operatorname{Tr}_B (B(t) B^\dagger(t_1) \rho_B(0)) \tilde{\rho}_s(t) c^\dagger(t_1)}_{\text{I.}} + \gamma \underbrace{\int_{-\infty}^t dt_1 c^\dagger(t_1) \operatorname{Tr}_B (B^\dagger(t) B(t_1) \rho_B(0)) \tilde{\rho}_s(t_1) c(t)}_{\text{II.}} \end{aligned}$$

$$\text{I.} = \gamma \int_0^\infty d\tau \operatorname{Tr}_B (B(t) \tilde{B}(t-\tau) \rho_B(0)) c(t-\tau) \tilde{\rho}_s(t) c^\dagger(t)$$

$$= \gamma \left(\int_0^\infty d\tau \operatorname{Tr}_B (B(t) \tilde{B}(t-\tau) \rho_B(0)) e^{i\omega_s \tau} \right) c(t) \tilde{\rho}_s(t) c^\dagger(t)$$

$$\text{II.} = \tau \int_0^\infty d\tau \operatorname{Tr}_B (B^\dagger(t) B(t-\tau) \rho_B(0)) c^\dagger(t-\tau) \tilde{\rho}_s(t) c(t)$$

$$= \gamma \left(\int_0^\infty d\tau \operatorname{Tr}_B (B^\dagger(t) B(t-\tau) \rho_B(0)) e^{-i\omega_s \tau} \right) c^\dagger(t) \tilde{\rho}_s(t) c(t)$$

$\Rightarrow$ all together we get

$$\begin{aligned} &\gamma \left[c^\dagger(t) \tilde{\rho}_s(t) c(t) \cdot \left(\int_0^\infty d\tau \operatorname{Tr}_B (B^\dagger(t) B(t-\tau) \rho_B(0)) e^{-i\omega_s \tau} + \operatorname{Tr}_B (B^\dagger(t-\tau) B(t) \rho_B(0)) e^{i\omega_s \tau} \right) \right. \\ &\quad \left. + c(t) \tilde{\rho}_s(t) c^\dagger(t) \left(\int_0^\infty d\tau \operatorname{Tr}_B (B(t) \tilde{B}(t-\tau) \rho_B(0)) e^{i\omega_s \tau} + \operatorname{Tr}_B (B(t-\tau) B^\dagger(t) \rho_B(0)) e^{-i\omega_s \tau} \right) \right] \end{aligned}$$

A_+

A_-

14. Stroboscopic master equation (2P)

Consider a cavity QED system as discussed in the lecture in Sec. 3.3, where Rydberg atoms with states $|g\rangle$ and $|e\rangle$ are interacting **resonantly** with a **single microwave** mode. During the time of interaction the Hamiltonian is

$$H_{\text{int}} = \frac{\hbar g}{2} (a\sigma_+ + a^\dagger\sigma_-). \quad (5)$$

Assume that at a time t the cavity is an arbitrary pure state $|\psi_c(t)\rangle$ and a Rydberg atom in state $|g\rangle$ is sent toward the cavity. Therefore, right before the interaction the total state of the system is $|\Psi(t)\rangle = |\psi_c(t)\rangle \otimes |g\rangle$.

- (a) Assume that the Rydberg atom is very fast such that the parameter $\varepsilon = \tau g/2 \ll 1$ is small, but finite. Evaluate the state after the interaction,

$$|\Psi(t+\tau)\rangle = e^{-iH_{\text{int}}\tau/\hbar} |\Psi(t)\rangle, \quad (6)$$

up to order ε^2 .

$$(-i)^2 = -i \cdot -i = i^2 = -1$$

$$\begin{aligned} e^{-iH_{\text{int}}\tau/\hbar} &\approx 1 + (-iH_{\text{int}}\tau/\hbar) + \frac{1}{2}(-iH_{\text{int}}\tau/\hbar)^2 + \mathcal{O}(\varepsilon^3) \\ &= 1 - i\varepsilon(a\sigma_+ + a^\dagger\sigma_-) - \frac{\varepsilon^2}{2}(a\sigma_+ + a^\dagger\sigma_-)^2 \\ &= 1 - i\varepsilon(a\sigma_+ + a^\dagger\sigma_-) - \frac{\varepsilon^2}{2}(\underbrace{a^2\sigma_+^2}_{\rightarrow 0} + \underbrace{a^{\dagger 2}\sigma_-^2}_{\rightarrow 0} + a a^\dagger \sigma_+ \sigma_- + a^\dagger a \sigma_- \sigma_+) \\ &= 1 - i\varepsilon(a\sigma_+ + a^\dagger\sigma_-) - \frac{\varepsilon^2}{2}(\hat{n}|g\rangle\langle g| + \underbrace{a a^\dagger}_{=1+a} |e\rangle\langle e|) \end{aligned}$$

$$\sigma_+\sigma_- = |e\rangle\langle g|g\rangle\langle e| = |e\rangle\langle e|$$

$$\sigma_-\sigma_+ = |g\rangle\langle e|e\rangle\langle g| = |g\rangle\langle g|$$

$$a a^\dagger = 1 + a^\dagger a$$

$$\begin{aligned} e^{-iH_{\text{int}}\tau/\hbar} |\Psi\rangle &= e^{-i\varepsilon} |g\rangle \otimes |\psi_c(t)\rangle \\ &= |g\rangle |\psi_c(t)\rangle - i\varepsilon |e\rangle a |\psi_c(t)\rangle - \frac{\varepsilon^2}{2} |g\rangle \hat{n} |\psi_c(t)\rangle \end{aligned}$$

$$\begin{aligned} |\psi_c(t+\varepsilon)\rangle &= \sum c_n |n\rangle \rightarrow a |\psi_c(t)\rangle = \sum \sqrt{n} c_n |n-1\rangle \\ &\rightarrow \hat{n} |\psi_c(t)\rangle = \sum n c_n |n\rangle \end{aligned}$$

b)

$$\rho_c = |\psi_c(t+\varepsilon)\rangle\langle\psi_c(t+\varepsilon)|$$

$$\begin{aligned} &(|g\rangle\langle g| |\psi_c(t)\rangle\langle\psi_c(t)| - i\varepsilon |e\rangle\langle e| a |\psi_c(t)\rangle\langle\psi_c(t)| - \frac{\varepsilon^2}{2} |g\rangle\langle g| \hat{n} |\psi_c(t)\rangle\langle\psi_c(t)|) (\langle g| \langle\psi_c(t)| + i\varepsilon \langle e| \langle\psi_c(t)| a^\dagger - \frac{\varepsilon^2}{2} \langle g| \langle\psi_c(t)| \hat{n}) \\ &= |g\rangle\langle g| \otimes |\psi_c(t)\rangle\langle\psi_c(t)| + i\varepsilon |g\rangle\langle e| |\psi_c(t)\rangle\langle\psi_c(t)| a^\dagger - \frac{\varepsilon^2}{2} |g\rangle\langle g| |\psi_c(t)\rangle\langle\psi_c(t)| \hat{n} \\ &\quad - i\varepsilon |e\rangle\langle g| a |\psi_c(t)\rangle\langle\psi_c(t)| + \varepsilon^2 |e\rangle\langle e| a |\psi_c(t)\rangle\langle\psi_c(t)| a^\dagger - \frac{\varepsilon^2}{2} |g\rangle\langle g| \hat{n} |\psi_c(t)\rangle\langle\psi_c(t)| + \mathcal{O}(\varepsilon^3) \\ \rightarrow \text{Tr}_{\text{atom}}(\rho_c) &= |\psi_c(t)\rangle\langle\psi_c(t)| - \frac{\varepsilon^2}{2} |\psi_c(t)\rangle\langle\psi_c(t)| \hat{n} + \varepsilon^2 a |\psi_c(t)\rangle\langle\psi_c(t)| a^\dagger - \frac{\varepsilon^2}{2} \hat{n} |\psi_c(t)\rangle\langle\psi_c(t)| \end{aligned}$$

c)

$$\begin{aligned} \rho_c(t+\varepsilon) - \rho_c(t) &= |\psi_c(t)\rangle\langle\psi_c(t)| - (|\psi_c(t)\rangle\langle\psi_c(t)| - \frac{\varepsilon^2}{2} |\psi_c(t)\rangle\langle\psi_c(t)| \hat{n} + \varepsilon^2 a |\psi_c(t)\rangle\langle\psi_c(t)| a^\dagger - \frac{\varepsilon^2}{2} \hat{n} |\psi_c(t)\rangle\langle\psi_c(t)|) \\ &= \varepsilon^2 (|\psi_c(t)\rangle\langle\psi_c(t)| \hat{n} - a |\psi_c(t)\rangle\langle\psi_c(t)| a^\dagger - \frac{1}{2} \hat{n} |\psi_c(t)\rangle\langle\psi_c(t)|) \\ \frac{\dots}{\varepsilon} &= \varepsilon \cdot \frac{g}{2} (\dots) \quad \text{if we use } n = a^\dagger a, \quad c^\dagger = a^\dagger, c = \hat{a} \end{aligned}$$

this is equal to the Lindblad equation with $\Gamma = \frac{\varepsilon \cdot g}{2}$,

d) if we start in $|e\rangle$, then

$$= 1 - i\varepsilon(a\sigma_+ + a^\dagger\sigma_-) - \frac{\varepsilon^2}{2}(\hat{n}|g\rangle\langle g| + a a^\dagger|e\rangle\langle e|)$$

$$e^{-i\varepsilon H}|\psi\rangle = e^{-i\varepsilon H}(|e\rangle \otimes |\psi_c(t)\rangle)$$

$$= |e\rangle \otimes |\psi_c(t)\rangle - i\varepsilon a^\dagger|g\rangle|\psi_c(t)\rangle - \frac{\varepsilon^2}{2}a a^\dagger|e\rangle|\psi_c(t)\rangle$$

$$= |e\rangle \otimes |\psi_c(t)\rangle - i\varepsilon a^\dagger|g\rangle|\psi_c(t)\rangle - \frac{\varepsilon^2}{2}\hat{n}|e\rangle|\psi_c(t)\rangle$$

$\Rightarrow$ in this case, we'd get the same result but with $a^\dagger \leftrightarrow a$:

$$\rightarrow \mathcal{P}_A(t) = |\psi_c(t)\rangle\langle\psi_c(t)| + \varepsilon^2 a^\dagger|\psi_c(t)\rangle\langle\psi_c(t)|a - \frac{\varepsilon^2}{2}\hat{n}|\psi_c(t)\rangle\langle\psi_c(t)|$$

$\Rightarrow \mathcal{P}(t+\varepsilon) - \mathcal{P}_A(t)$ then gives the same equation as before, just with $a \leftrightarrow a^\dagger$ swapped

$$[H, \rho] = H_{\text{int}}|e\rangle\langle e| - |e\rangle\langle e|H_{\text{int}}$$

$$= a^\dagger|g\rangle\langle g| - |g\rangle\langle g|a$$

$$H_{\text{int}} = \frac{\hbar g}{2}(a\sigma_+ + a^\dagger\sigma_-).$$

$\rightarrow \text{tr}_{\text{atom}}(\dots) = 0 \rightarrow$ the additional term can't be the $[H, \rho]$ term